Modeling Fantasy Football Player Performance with an Emphasis on Rookie Players

Kush Lalwani[#1], Trevor Bunch[#2]

[#1]*College of Engineering,*
The Pennsylvania State University,
United States of America

[#2]*College of Information Science and Technology,*
The Pennsylvania State University,
United States of America

[1]kgl5163@psu.edu

[2]twb5545@psu.edu

Abstract— This document will cover our team's research and work into developing a data centric approach to project an NFL player's fantasy football points for an upcoming season, with a focus on improving volatile rookie player prediction. Fantasy football prediction is a complex task due to the unpredictability of player performance, noisy football game factors like team strategy or opponent, and external sources like weather or injury. Our work sourced years of historical player data and introduced a novel approach to giving rookies historical data via college statistics. With the data, alongside processing and feature engineering, we developed machine learning models to project fantasy points for NFL players during the 2024 NFL season.

Our developed methodology used five (5) regression-based models like Random Forest and Gradient Boosting on each important fantasy football positional group, QB, RB, WR, TE, to predict their points. Our work utilized training and testing data with hyperparameter tuning and backwards feature selection to find the best model with the lowest RMSE for each position, where our validation data would be run through to get predictions. Our findings show that our novelty ideas regarding rookies help improve their predictive accuracy, with low MAE ranks. In addition, they show in general the complexities with predicting a noisy subject, like fantasy football, with erratic RMSEs. Despite this, we were able to produce player rankings in respect to other players that are realistic.

Keywords— Fantasy Football, NFL Prediction, Regression, Machine Learning, Rookies, Point Projection, Forecasting

I. INTRODUCTION

Fantasy football is an online user game where players construct rosters of real NFL players and go head-to-head against other players, seeing whose roster can score more points. Points are derived from real life NFL players' performances, with conversion of their box score stats into fantasy points. There are multiple conversions or scoring settings available, but our work focuses on the Point Per Reception (PPR) format. In addition to the diversity of scoring settings in fantasy football, there are many ways for users to build their roster - we will be focusing on the common pre-season draft approach, where users draft lineups against each other. The last customizable feature of fantasy football is what position players can make up rosters, and for the purpose of our work we focused on the four highest scoring: Quarterback (QB), Running back (RB), Wide Receiver (WR), and Tight End (TE).

Fantasy football has experienced significant increased user activity and popularity over recent years, which turned it into a multi-billion-dollar industry with many sports companies and betting sites getting involved. The surge of activity has brought new and casual users to the game who aren't as experienced with the how the game is played and the best strategies to win, which forms a problem. These users need additional insights to help form their roster during the pre-season draft process and gain a competitive edge over their league mates. This is a challenge due to the noisiness and unpredictability of fantasy football and the NFL.

Through our research we found that using data and machine learning can help overcome this issue, as many companies/sites utilize this to form their own pre-season projection rankings to help. Using data-driven solutions can give the best projections and insights to overcome the noise; however, current solutions perform poorly when predicting rookies.

Rookies are a vital part of fantasy football, as typically a few of them rank very highly within their position by the end of the year. They have no historical data, which makes them extremely hard to predict. Due to the unpredictability, rookies rank lower in projections, but can exceed them greatly, providing users with a "steal" in their fantasy draft.

Our innovation and novelty approach involved

improving rookie predictions by giving them historical data to reduce their volatility. We did this by adding in their college statistics from their last year before they got drafted into the NFL. We also accounted for the competitive imbalances seen in college football (which leads to skewed statistics) by including in team rank, strength of schedule and conference.

Additionally, to address the problem of overall noise in fantasy football predictions, we have a few further innovations. For predicting non-rookie players, we included additional features that go beyond normal box-score statistics. Included is a fantasy strength of schedule feature that measures how well a player's opponent defends against their position. This feature considers how easy or hard the NFL player's opponents are for the upcoming season, which can affect their points. Additionally, we have a feature that measures how good an overall NFL team is. Better NFL teams typically perform better, score more and have higher fantasy points. It is important to consider these additional factors, and not just box score statistics when predicting fantasy points.

With the motivation, problem, and our novelty approaches in mind, we have multiple goals:

Goal #1: Predict 2024 NFL non-rookie player fantasy points using their historical data and our novelty features to then rank them.

Goal #2: Predict 2024 NFL rookie player fantasy points using their college data and our novel features to rank them.

Goal #3: Be able to predict rookies more accurately than past research.

We are predicting the 2024 NFL season because it is the most recent fully completed season where there are end rankings we can validate and compare our results with.

These goals will help produce insightful rankings to be used by users (specifically casual ones) to help them in their pre-season draft. A data and modeling approach to this problem can give insights and help put basis and confidence into pre-season fantasy draft decisions, especially considering the noise of the topic and the possible inexperience of the user.

Our purpose is to overcome the challenge of noisy and erratic NFL data, especially with rookies, to predict fantasy points for an NFL player. Then based on the predicted fantasy points, a pre-season ranking list can be made. By looking at the rankings list, a drafter can simply see who is ranked higher and should be drafted in their spot. We aim for our rankings list and rookie predictions to be better than other popular rankings like ESPN or Sleeper due to our innovations.

II. LITERATURE REVIEW

Previous research in fantasy football projection has taken many different approaches to predicting player performance. Over time, the field has evolved from simple statistical regression models to more complex machine learning algorithms that incorporate features beyond traditional "box score" statistics. While these data-driven decision models show strong evidence that they can outperform human intuition, several key limitations remain. One persistent challenge lies in the lack of accurate fantasy forecasting for rookie players, who have little to no professional data available. The following literature review examines prior work across three major areas: statistical regression models using structured NFL game statistics, multimodal systems that integrate unstructured-contextual information, and optimization-based approaches for lineup selection. Together, these studies form the foundation on which our research is built. We aim to bridge gaps in rookie player prediction and model interpretability while incorporating contextual features into our fantasy football projection model.

The most relevant study in this area [1] focused on developing a machine learning framework to predict the top twelve fantasy players during the 2024 NFL season. The researchers separated each of the previously mentioned position groups (QB, RB, WR, TE) including structured data in the individual groups. The data was sourced from *Pro Football Reference,* which included metrics such as passing yards, touchdowns, and other performance indicators. The task was treated as a regression problem, testing several algorithms including Random Forest, Linear Regression, and Gradient Boosting. After developing the regression model, the researchers implemented contextual features such as injury status and schedule strength to better simulate real conditions and improve prediction accuracy. An R-squared value of 0.99 and a root mean squared error (RMSE) of 53.85 was achieved using a Random Forest model, showing that non-statistical factors do play a role in overall player score. Unfortunately, this research excluded rookie players due to the lack of historical NFL data, limiting the scope. Additionally, the researchers were unable to fully validate their findings because the 2024 season had not been completed at the time of publication. This project provides a relevant guide for our project to follow, which includes testing multiple models trained on both statistical and non-quantitative statistics. Additionally, the study's approach of displaying the final projections as a table of the top twelve players by position is something we look to adapt. We aim to extend this research by incorporating college-level data for rookie players (omitted entirely in [1]) to create more inclusive and realistic projections.

Another significant contribution to fantasy football forecasting comes from research that integrates unstructured media data and contextual information into predictive modeling. [2] expanded this idea by developing a multimodal deep learning model that combines structured NFL statistics with a natural language processing (NLP) of sports media data and player sentiment. This process began with over 100 GB of sports text and multimedia content collected from ESPN, news outlets, and social media. This data was labelled with 13 entity types depending on the contents (injury, performance, etc.). This data was then fed into doc2vec and transformer-based models to encode sentiment. Then a deep neural network (98 layers) using the sentiment output and raw stats was used to classify players into four categories: Boom, Bust, Hidden Injury, and Meaningful minutes. This methodology combined

structured ESPN statistics with NLP-derived features to create an ensemble regression model. After developing the final model, AI Fairness 360 was implemented to detect and correct any bias present in predictions across teams or player demographics. This complex methodology was able to create a model that achieved 88% prediction accuracy within ten points of actual fantasy outcomes. The sentiment classifier reached 81% accuracy, confirming that the unstructured context improves prediction quality. We look to incorporate some aspects of this research into our own. While we would look to incorporate the NLP portion, we do not have access to the computational power that IBM has. Instead, we will look at including other simpler media-based features. For example, we look to include preseason rankings for each team, which have been published by ESPN journalists before every NFL season. We are also looking to include the strength of schedule, which will be computed by averaging journalist-produced matchup difficulty ratings.

As prediction models become more advanced, research began shifting towards applying these forecasts to strategic decision-making. This next area of study focuses on how optimized lineup creation can be achieved through a combination of machine learning and mathematical programming. [3] looked specifically at the domain of daily fantasy sports (DFS). DFS is a special type of fantasy competition where the competitions are over a shorter period of time than the typical season-long fantasy competition. They are typically constrained to a week or even a day period, and most importantly, DFS allows the same player to be present on multiple users' rosters. The main restriction is a salary cap or set budget that all selected players must be under. The researchers' purpose through this paper was to combine both machine learning for predicting players' scores and linear programming to optimize the selected lineups. They started with 2018 season data from popular fantasy companies like DraftKings and ESPN. From this data, they trained a two-layer neural network, which took in 43 input features and output a single number: the projected fantasy score. These projections were then fed into a mixed integer linear program (MILP). This was used to maximize the number of points for an entire roster under the budget constraint. They aimed to maximize points using a typical nine-player fantasy roster. After obtaining the final predictions and optimized lineups, they benchmarked the model against other randomized lineups and real-user lineups. The model was able to outperform the randomized lineups and score in the 31st percentile against user-generated lineups. The predictions significantly declined as the season progressed, likely due to injuries and dynamic team changes- once again highlighting the importance of context-based features in predictive modeling. Through this study, the researchers confirmed the importance of incorporating optimization into fantasy sports. For our paper, we will look to incorporate this fusion of prediction and optimization. In our case, regression will be used to project a player's points over the entire course of the season, instead of only a DFS length competition. Optimization will be factored into team selection during a typical fantasy team's start-up draft. It will be used to optimize the player choices with constraints on roster space and the players still yet to be selected.

Collectively, these studies demonstrate the evolution that fantasy football analytics; from simple regression models to advanced multimodal systems and optimization methods. Each of these papers addresses different aspects of prediction and decision making when it comes to fantasy football. Together, they show how the integration of machine learning, context-based data, and optimization has significantly improved projection accuracy and data-driven lineup construction. However, they also highlight a common problem with fantasy football prediction; while accuracy and real-world applicability have improved, the challenges of modeling rookie players and interpretability remain unsolved. Prior approaches either exclude rookies due to missing historical data or rely on complex models and data processing that are difficult to scale and reproduce. This highlights the need for a framework that combines the strengths of these studies while remaining logistically practical, interpretable, and inclusive of both rookie and veteran players.

To further build upon the ideas presented in these studies, our research aims to address one of the most glaring gaps in fantasy football prediction models: the lack of effective methods for evaluating rookie players. By integrating college-level statistics to simulate historical performance for rookie players and including other context-based features, our approach extends the viability of the previous models in the fantasy football scope. This proposed framework provides a perfect balance between predictive accuracy and interpretability, allowing for more inclusive and practical preseason fantasy projections that can be easily understood by users.

III. IMPLEMENTATION

The implementation of our prediction model follows a structured end-to-end workflow that closely follows the data science pipeline. It begins with collection of all raw data. Following this step is intensive data cleaning, standardization, and feature engineering. These steps provide two separate datasets that will be used as inputs to model the fantasy performance of veterans and rookies. After the data was prepared, they were fed into multiple machine learning models for each player group, to account for different performance histories. In addition to model training, hyperparameter tuning and feature selection was implemented to find the best possible model for each sub-group. The following section describes the data acquisition phase, which serves as the foundation for this modeling pipeline.

A. Data Acquisition

The first stage of putting this project into action was data acquisition. This phase focused on collecting any sort of data that we thought would be relevant to predicting fantasy football scores for both NFL veterans and rookie players. The goal was to build parallel datasets for both groups of players, so that both groups could be fairly and consistently modeled. This is because each of the groups rely on different forms of data (NFL vs. college). The following section will discuss the entire process of acquiring such relevant information.

Before searching for such data, it was important to first think about the scope of the problem we are looking to address,

which was "How can we accurately model PPR fantasy football points for a player?". The most obvious features to consider are the raw statistics from the previous season. So for such data, we had to consider the timeline for our project. We wanted to be able to validate the results of our model, which was a major flaw of [1], so we decided to choose our projections of the 2024 season as our validation set. This means that statistics used to predict the 2024 season would come from the previous year, 2023. This approach was selected to avoid any sort of data leakage to the model which could cause an artificially positive result. We decided that five seasons worth of training data should be sufficient to derive any sort of relationships between predictors and the target variable. Now that the timeline for the project was set (using season stats from 2019-2023), it was now time to worry about how to process such data. When going about this data processing, it was very important to work on both the rookie and veteran datasets in parallel, as not to favor one over the other, potentially tainting our results.

As mentioned previously, we used NFL season data from 2019 to 2023 seasons. This data consisted of basic statistics such as passing yards, touchdowns, etc. In parallel, we collected college football data from the same timeframe as the NFL data. The college football statistics were broken into three separate tables: one table just for passing statistics, another table for rushing, and another table for receiving statistics. All of this data was easily available from *Pro Football Reference*, which had all data readily available in csv format. This is the same approach to collection that [1] used; however, they did use a much wider time period from 2009 to 2023.

Connecting to [2], we also wanted to implement a few potential features that were based on sports media and journalism. While we may not have the computational power of IBM to process gigabytes of multimedia, we came to a simple approach to implementing some of the context-based features. A dataset was manually constructed based on ESPN's preseason rankings of each NFL team. This table ranked NFL teams from 1 to 32 based on sports journalist's thoughts about each team. This data was collected for years 2020 to 2024, as it was a prediction done before each season had started.

Another context-based feature that we looked to implement was a strength-of-schedule (SoS) for each team. While it was not possible for us to find any ready-made files that had this data, we felt it was too much of an important feature to omit; so we decided to implement our own.

Sleeper.com is a popular website for fantasy football, which has many different types of statistics readily available to users. One of these features is a matchup rating for players against teams based on the player's position (e.g. The Philadelphia Eagles may be an easy match-up for opposing TEs but can be hard matchup for opposing RBs). The matchups are categorized into one of four levels: dark green, an easy matchup; light green, a moderately easy matchup; orange, medium level; and red, a difficult game. These ratings are determined by fantasy football experts and sports researchers before the season starts.

Additionally, a file containing the entire NFL schedule from our selected years was downloaded.

Both the *Sleeper.com* matchup rankings combined with the NFL schedule provide the perfect base for our SoS feature to be implemented. We will go more into depth on the steps of building this feature in Feature Engineering [Section III-C].

For the rookies, we do need a bit more help to bridge the gap between college and the NFL. To do this we researched a couple of new features for the college players. After exploring *Pro Football Reference* we determined the draft order would be a great indicator as to if a player performs well or not. The NFL draft is where teams are able to strengthen their teams by selecting college players to represent them in the NFL. Typically, the better players are selected earlier in the draft, which could be an indicator of the player's potential for success in the NFL.

We also looked to address the issue of not all colleges having equal performance. To account for the imbalances between colleges and conferences, we acquired data that ranked each of the college conferences for every season in question. The same was done for individual schools, ranking every college from 1 to about 130.

Together, these sources provide the foundation for constructing two aligned datasets: One for veterans and one for rookies. The cleaning, standardization, and merging of this data will be described in detail in the following section.

B. Data Cleaning and Standardization

The next step towards our final product is to get all of our data into a format that is consistent and model ready. This phase ensures that all identifiers, variables, and formats match across the derived datasets. All work done in this section needs to be carried out for both the veteran NFL players and the rookie NFL players. This does lead to a difference in methodology for both groups of players, as their data is formatted differently

With the veteran players, each season's data was loaded into a Jupyter Notebook. First we need to identify how to create a primary key for joining future data. Using a combination of player name, year, and team was used as the key for joining based on the data being joined. To get this key, a year variable was added to each player's statistics that corresponds to the year that the statistics were from.

Additionally, some issues with the original data source were cleaned up here. The data contains two-level headers and some special characters around the player names. These issues were solved by renaming the columns and stripping any special characters from the players' names. This included suffixes like Jr. or III. in a player's name.

Then any places where NAs existed were carefully examined. Most of them appeared in the statistics relating to two-point-conversion. These were likely left as Nas because they are not very common in the NFL, leading to sparsity in the dataset. Any missing values in this column were filled with zeroes.

Another column that brought about missing values was the Fantasy_PPR column, which stores the players fantasy score for the corresponding season. Filling in this missing data was quite simple, as we only needed to use the basic PPR format formula for scoring.

$$Fantasy_PPR = 0.04(\text{Passing_Yds}) + 4(\text{Passing_TD}) - 2(\text{Passing_TD}) + 0.1(\text{Rushing_Yds}) + 6(\text{Rushing_TD}) + 0.1(\text{Receiving_Yds}) + \text{Receptions} + 6(\text{Receiving_TD}) - 2(\text{Fumbles_Lost}) + 2(2\text{PM}) + 2(2\text{PP})$$

Any remaining NAs were due to a player simply not having statistics available. So for the few cases where this occurs, the observations were just removed.

As a final step for the veteran players, we needed to create the target column, which is what the model will aim to predict. To do this, we use the entire merged dataset- which consists of all the years of data- and shift the column with the fantasy score back by a year (e.g. a player's 2023 stats would have their 2024 fantasy score attached as the target). We also shifted the team corresponding to the next season back as a player might be on a new team which is something we want to consider.

For the rookie players, the cleaning approach started with the NFL draft selections for the corresponding year. After adding a new column that represented the year of the draft, which would also be the year of the rookie's first NFL season. Next, we ensured that we only selected players that are relevant to fantasy football; meaning that we removed all defensive, special teams, and offensive-line players from our dataset. This is because while real NFL teams may need these players on their teams, they are not traditionally involved in fantasy football.

After this, we selected only relevant columns for our analysis such as pick number and college. We needed to rename many of the colleges in this draft dataset, as they use abbreviations or names that are not present in the statistics dataset (e.g. TCU vs. Texas Christian).

Since the draft data was completed, we will now work on the collegiate statistics for each of these drafted players. To do this we looked at the three datasets previously mentioned (passing, rushing, and receiving statistics), and joined them using player name, college team, and draft year as the keys. Draft year was an important feature to use for joining; this is because the same player may have multiple seasons in college, using the draft year ensures that we have the most recent collegiate season for each player.

After this merging of multiple datasets, we checked for NAs. Two cases of missing values were easily solved. Players Ja'Marr Chase and Nico Collins took a one-year break from football before entering the NFL Draft in 2021. For this we just incremented the year key for their college stats to include the most recent year, modeling their 2019 stats as 2020.

The other NAs were present because some colleges were not available in the dataset. There was no easy solution to this issue, so we needed to remove them because there were no statistics to be used for predictions.

Then, we followed the same approach as the veterans for attaching the target column to the rookie's college statistics.

Both the veterans and the rookie's data set required intensive cleaning and standardization to ensure that it will be in a model-ready format. The rookies did require significantly more preprocessing than the other dataset due to the nature of the data. Now that the data is in a clean format, it is now possible to create new predictors to the model to more precisely forecast the players' fantasy football score.

C. Feature Engineering

This step, feature engineering, is one of the most important steps of our modeling pipeline. It will allow us to add more predictive power to our model by creating new and unique predictors. For this step we looked to focus primarily on the multi-media data that was mentioned in [2]. This is because we already have lots of in-game statistics. We also want to be able to focus on data that has an impact but is often overlooked.

We wanted to start this process with the SoS metric that was previously mentioned as it was the most computationally intensive feature.

To do so, the dataset containing the NFL schedules was parsed to transform it into a new format. The new format used each of the 32 NFL teams as a primary key, and the following columns correspond to each opponent that the team faced during the season. This approach ignored bye-weeks, focusing only on each of the 16 or 17 teams that were played each season.

This is where our *Sleeper.com* matchups-by-position data comes in. We mapped each of the colors used to denote the matchup difficulty to a number from 1-4, which 1 being the most difficult and 4 being the easier matchups.

Using a complex series of joins on the reformatted NFL schedule data, we were able to construct a dataset that contained the each of the 32 NFL teams and their defensive matchup rankings by position per year. To simplify, the dataset contained 128 rows of data per year (four per team); all opponent's defensive strength against each of the four main positions (QB, RB, WR, TE).

The columns of this new table were aggregated to create an average SoS by position for each team. This was on a scale of 1-4, so to make it more interpretable we recomputed it as a z-score. This meant that harder schedules are denoted as a high magnitude negative number and easier schedules are a high magnitude positive number.

We then joined this to each of the players in the dataset, both rookies and veterans, using team, year, and position as the join variables.

Next, we looked at the preseason rankings from ESPN that were collected at the start. These were rankings of teams before each season started from 1-32 with 1 being the strongest team and 32 the weakest.

This data was attached to the main data frames using the year and the team name.

At this point there was nothing more that we needed to do for the veteran players. So we exported their completed dataset to a new file that is ready for model training. The rookies, however, had some more work to be done.

As mentioned previously, we already have engineered features for SoS and preseason rankings for the rookies' NFL teams. But there is more that needs to be done to 'bridge the gap' between collegiate football and professional football. As mentioned in [Section III-A] we are going to add college and conference rankings to help

correct the imbalances between colleges and conferences.

The first of the college's specific multi-media features was the conference ranking. This was important to add to the model because not all college conferences are equal. For example, the Southeastern Conference (SEC) is widely regarded as the best conference in college football. Because of these imbalances, we integrated a ranking for each of the 11 (now 10) college football conferences.

This was also simple to implement, joining our rankings dataset with the rookie data frame on the variables: year and conference name.

The next context-based feature we are going to implement is team ranking. This is also an easy feature to implement based on the data we already have downloaded. This data contains rankings of most of the college football teams ranked from 1 to about 130.

This was again a simple join on the year and college team name.

Now that the rookie dataset has all the feature that we require for modeling, we are now able to export this dataset as complete.

The veteran player dataset contains 2554 player observations with 27 columns across all four fantasy relevant positions. For the rookie dataset, we have a smaller dataset of 371 observations and 26 columns to partially bridge the gap to the professional players.

Both datasets will be fed into separate machine learning pipelines respectively, one for rookies and one for veterans. Also, each of the datasets will be split by position. This approach will be detailed further in the following sections.

D. Training, Test, and Validation Sets

Continuing along with the data science pipeline, we had to split our datasets into train, test, and validation sets for modeling. First, we wanted to separate our validation data from each and then have it unseen until the end. For the veteran NFL player data, the 2023 season was made the validation set because we are predicting the next season, 2024. Similarly, the rookie player data's validation set became rookies drafted in 2024, since we are predicting the fantasy points for that season. The remaining veteran data consisted of the years 2019-2022 and the remaining rookie data consisted of players drafted in the years 2020-2023. Note the difference in years is due to the NFL draft occurring after the NFL season.

From both of the remaining datasets, each were broken up by the position groups of interest: QB, RB, WR, TE. This means we had veteran QB, RB, WR, TE sets and rookie QB, RB, WR, TE sets. This is important because we want to find the best model and features for each position, as each have their individual uniqueness. The target column was then separated from each positional set and made into its own data.

Additionally, each of the validation sets were broken down into the four positions of interest.

We have input (X) and target (y) data for each position for both veterans and rookies. A train/test split of 80/20 was used for each positional set for both vets and rookies to form the training and testing data. This data will be fed into the models to determine the optimal model, hyperparameters, and features for predicting fantasy points.

E. Machine Learning Methods and Tuning

Predicting fantasy points is a regression problem, and by using our knowledge and research from the literature review, we determined five models to be looked at. The models are Random Forest, Lasso, Neural Network, Support Vector Regression, and Gradient Boosting (XGBoost). Important aspects for our models are hyperparameter tuning and feature selection to get the best possible predictions.

Our hyperparameter tuning consisted of using a 3-fold cross validation grid search to find the best possible combination of hyperparameter values for each model. Each model in our pipeline has common hyperparameters and possible associated values for each, that are looked at during the grid search. The grid search is fit with the training data for the position being looked at in the moment (e.g. veteran QB, rookie WR). This again is because we want to find the best model with the best hyperparameters for each position. The best model and hyperparameters are determined using RMSE. The grid search first determines which combination of hyperparameters gives the lowest RMSE for each model. This gives the best possible version of all 5 models for each position. The models are then sorted to find the one with the lowest RMSE, which is the best model for the position. The best model found for the position then undergoes feature selection.

Our feature selection focused on using backwards feature selection to iteratively remove the least important features until the RMSE stopped improving. This is key for predictions as certain stats pertain more to certain positions. For example, a Quarterback relies mainly on passing and sometimes rushing stats, but not receiving stats. It can help our predictions by pruning possible irrelevant stats for each position set. From the best model and hyperparameters for the position, backwards feature selection is performed to determine feature selection and the best features for each position.

After determining the best model with its best combination of hyperparameters and the best features from the data, testing predictions are made. The positional training data with its best features is fit onto the best model and then the testing data with the best features is predicted onto the best model. This gives us an idea of how predictions look and how we think our validation data will perform.

The purpose of all the modeling, tuning, and selection is to ensure we have the best possible design in place for our validation data to give the best predictions for 2024 fantasy points. After finalizing the best models, hyperparameters, and features for veteran and rookie positions, we made predictions. The positional validation data for vets and rookies is used on its correlating best design to make 2024 fantasy point predictions. From the fantasy point predictions, we made fantasy positional rankings for both the veterans and rookies, but also an overall combined veteran ranking and overall combined rookie ranking. Lastly, we computed the RMSE, MAE, and R^2 for the predicted 2024 points of a player versus their actual 2024

points, along with the MAE for their predicted versus actual ranking.

IV. NOVELTY

The novelty implementation regarding our project revolves around the focus of predicting rookie NFL players. Rookies are a vital part of fantasy football, with typically some of them ranking towards the top performing players after their first season. However, our parent paper [1] omitted to predict them due to them not having any historical data playing in the NFL. Additionally, found in our research [2] that even when papers attempted to predict rookies, the results were typically poor. This motivates our novelty approach: we want to bridge the gap between a rookie's college career and NFL and give them historical data to predict their rookie season accurately. Compared to our parent paper, we are predicting rookies and hope to do it more successfully than others attempted, and we believe this because of our novelty data and feature engineering for our rookie models. We will have a pipeline made for predicting veteran NFL player fantasy points, as per our parent paper, but we build on that by creating a separate pipeline for predicting rookie player points. Additionally for our novelty, we engineered context-based features into our data that can help capture the unpredictability of fantasy football beyond just basic box-score statistics.

To support this expanded scope, our novel approach begins at the data level. Since rookie players cannot rely on historical NFL performance, we designed a data pipeline that reconstructs NFL-equivalent information using their college careers, draft context, and team environment. This required integrating several unique datasets, feature engineering, and developing contextual variables. The following text outlines the key innovations that we made to enable the rookie model to be comparable to our veteran model and address the limitations identified in our referenced studies.

Our first novel implementation was the integration of college and NFL datasets to create a continuous, comparable representation of player performance across the two very different levels of competition. Through this, we were also able to build an entire data pipeline for rookie projections that "converts" the college statistics into NFL-aligned inputs to the model. To do this we used the previously mentioned features such as draft information, college team, and college conference.

Another approach that was not taken in our reference papers was the creation of a novel SoS feature. Earlier research did implement a SoS metric based solely on general team ratings. Our approach aimed to be more granular, calculating the SoS by player position, allowing matchup difficulty to be captured more accurately. This specificity with the SoS metric allows for richer context and more relevant predictions.

Something else that built on from the previous work was our incorporation of the preseason rankings from sports journalists. For [1]'s integration of a team ranking was done so by using draft order, which is based on team record from the previous season. This approach fails to consider any new signings or trades a team has made in the offseason. Our approach of using preseason rankings helps to give more context around a team's offseason, which can allow for stronger predictions for both rookies and veterans.

On the model level, we had small innovations that differed from the parent paper [1]. The researchers manually selected important features for each position, whereas we determined feature importance in the model by backwards feature selection based on RMSE. This helps uncover potentially hidden relationships that can support predictions.

The novel contributions on the data level feed our innovations through the data science pipeline. It helps support the model level novelties which can help us achieve better predications for rookies. The overall goal was the emphasis on predicting rookie fantasy football points and rank, and doing it successfully. Looking into our findings, we believe our novelties helped us achieve this goal.

V. FINDINGS AND INSIGHTS

After implementing our methodology and novel ideas, we were able to produce results that predicted:

1. 2024 Veteran NFL player fantasy points
2. 2024 Veteran NFL player positional rank
3. 2024 Veteran NFL player overall rank
4. 2024 Rookie NFL player fantasy points
5. 2024 Rookie NFL player positional rank
6. 2024 Rookie NFL player overall rank

In addition, we applied RMSE, MAE, and R2 metrics to compare our predictions against our 2024 validation data which are the real-life results of the season.

A. 2024 Veteran QBs

The best model found for predicting veteran QBs is XGBoost with a test RMSE for the points around 77. The validation data was applied to this model and predicted QB fantasy points and ranks for 2024.

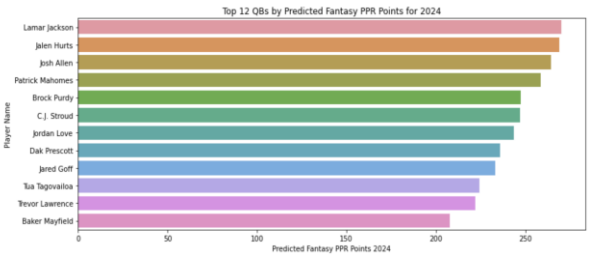

Fig 1. Top 12 Quarterbacks Predicted by our Model

We compared the real 2024 QB results to our predictions and got validation metrics:

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
QB	89.614877	68.691113	0.439528	12.875

Fig 2. Validation Metrics Calculated on QB Predictions

B. 2024 Veteran RBs

Neural networks performed best for predicting veteran RBs, sitting around a test RMSE of 60.

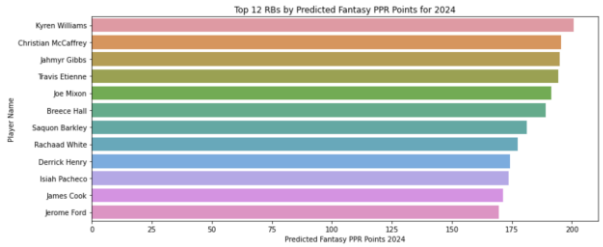

Fig 3. Top 12 Running Backs Predicted by our Model

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
RB	69.965786	51.998128	0.479905	17.542857

Fig 4. Validation Metrics Calculated on RB Predictions

C. 2024 Veteran WRs

The best performing model found in our pipeline for veteran WRs is Random Forest with an RMSE around 54. However, all other models were close, all within 3 RMSE points.

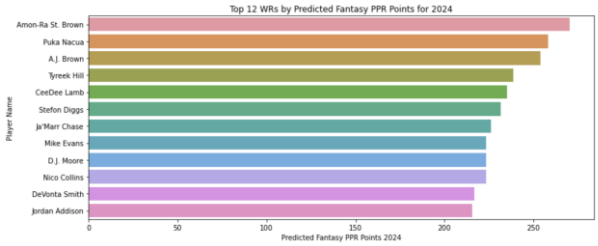

Fig 5. Top 12 Wide Receivers Predicted by our Model

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
WR	54.37794	40.937869	0.59237	29.336957

Fig 6. Validation Metrics Calculated on WR Predictions

D. 2024 Veteran TEs

The best model found for predicting TEs is XGBoost with an RMSE of 39. Additionally, all models did well for predicting TE's with the lowest attaining an RMSE of 43.

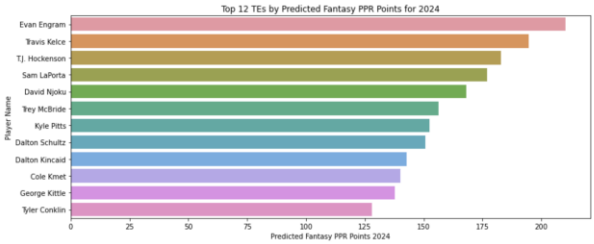

Fig 7. Top 12 Tight Ends Predicted by our Model

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
TE	41.68631	29.585584	0.529208	17.72973

Fig 8. Validation Metrics Calculated on TE Predictions

E. 2024 Overall Veterans

We were able to then combine all the positional rankings into an encompassing view of all the player's sorted by their predicted fantasy points.

Additionally, we were able to combine all their metrics found:

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
QB	89.614877	68.691113	0.439528	12.875000
RB	69.965786	51.998128	0.479905	17.542857
WR	54.377940	40.937869	0.592370	29.336957
TE	41.686310	29.585584	0.529208	17.729730

Fig 9. Combined Metrics Across on Positions

And a visualization was made to see how the metric values differed across position:

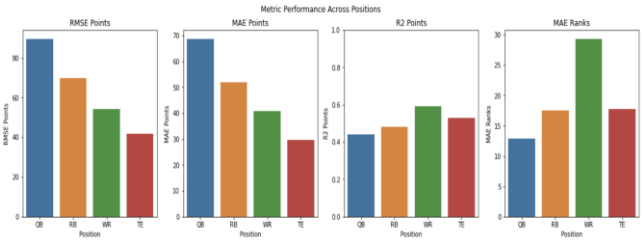

Fig 10. Visualization of Metrics Across Positions

F. 2024 Rookie QBs

For predicting rookie QBs in 2024, the best model in terms of RMSE is XGBoost around 68 points. However, all models were relatively close.

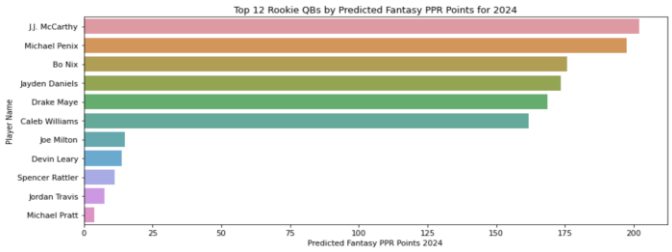

Fig 11. Top 12 Rookie Quarterbacks Predicted by our Model

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
QB	108.596973	78.7523	0.322525	2.363636

Fig 12. Validation Metrics Calculated on Rookie QB Predictions

G. 2024 Rookie RBs

LASSO performed the best looking at rookie RBs, with an RMSE hovering around 66.

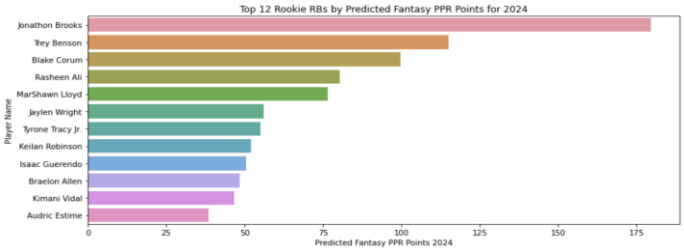

Fig 13. Top 12 Rookie Running Backs Predicted by our Model

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
RB	86.007986	63.766977	-0.595824	5.875

Fig 14. Validation Metrics Calculated on Rookie RB Predictions

H. 2024 Rookie WRs

XGBoost is the best performing model on rookie WRs. Interestingly, some of the models like SVR were quite bad, with RMSEs over 100 points higher than XGBoost's.

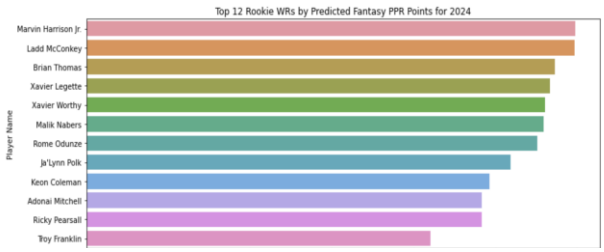

Fig 15. Top 12 Rookie Wide Receivers Predicted by our Model

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
WR	60.385889	45.765553	0.474048	5.529412

Fig 16. Validation Metrics Calculated on Rookie WR Predictions

I. 2024 Rookie TEs

Rookie TE's are best predicted by Neural Networks in our pipeline, with a low RMSE of 33.

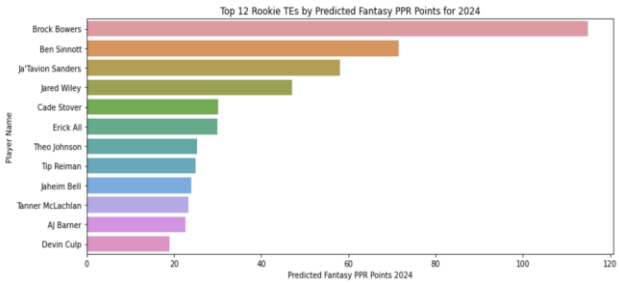

Fig 17. Top 12 Rookie Tight Ends Predicted by our Model

Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
TE	52.954076	36.535485	0.42538	2.833333

Fig 18. Validation Metrics Calculated on Rookie TE Predictions

J. 2024 Overall Rookies

All of the positional rookie predictions and rankings were concatenated into an overall rookie ranking regardless of position, descending by predicted fantasy points.

The combined metrics and metric visualizations were also made:

	Position	RMSE Points	MAE Points	R2 Points	MAE Ranks
0	QB	108.596973	78.752300	0.322525	2.363636
1	RB	86.007986	63.766977	-0.595824	5.875000
2	WR	60.385889	45.765553	0.474048	5.529412
3	TE	52.954076	36.535485	0.425380	2.833333

Fig 19. Combined Rookie Metrics Across on Positions

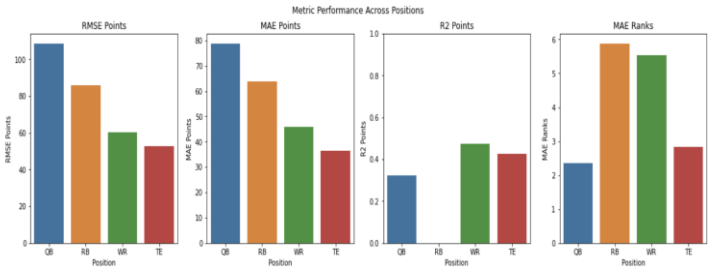

Fig 20. Visualization of Rookie Metrics Across Positions

K. Insights

Firstly, we can see our novel implementation worked well for predicting rookies. Compared to our parent paper, we were able to get point projections for rookies, which they failed to do. Building on this, the rookie projections, especially the ranks, were good as they had small MAE rank scores. This evidence helps support our innovation process. Giving the rookies historical data from their college playing time and additional information like NFL draft data, helped bridge the gap between the college and NFL ranks and give basis to a group that had none. When looking at the top predicted rookies at each position, all of them were very reasonable given the context and were decently accurate compared to real life results.

Additionally, we can see it is extremely difficult to predict fantasy points across all groups. While our rankings were good with lower MAE ranks across the board, the actual predicted points were far off from actual year end results, with higher RMSE points. This is due to a number of reasons. Injuries are nearly impossible to predict, and we saw numerous examples (e.g. Christian McCaffery) where a player was very highly projected, but injuries essentially derailed their entire season, and the fantasy points came nowhere close. Additionally, the NFL, sports, and fantasy are very noisy and unpredictable in general and not just injuries. There will always be examples of historically great players who will be very highly projected but simply don't meet expectation, like Tyreek Hill. Conversely, there will also be players who come out of nowhere and play amazing like Devon Achane. Essentially, the entire subject is extremely volatile and while we can do our best to put basis and data driven solutions into fantasy football it cannot capture all the unpredictability.

From our results we can derive insights from our models. XGBoost performed well across all positions, possibly due to its high predictive accuracy from its gradient boosting algorithm. Support Vector Regression did not achieve great results across our predictions, possibly due to the size and complexity of the data. XGBoost did a good job at uncovering the relationships in the data within the positions and helping our predicitons. Interestingly, when comparing a veteran position group to its rookie counterpart, only one position group had the same best performing model for both rookies and veterans: QBs. Both veteran and rookie QBs had XGBoost as their best predictive model, but every other position group had different models between vets and rookies. This could suggest different underlying relationships in the data we are maybe not considering and are not clear to us, which will require further investigation.

Next, when focusing on the position, there are more interesting results. Based on RMSE, we can say TEs are the easiest to predict. The second easiest to predict are WRs, the second hardest is RBs, and hardest is QBs. Logically, this does make sense. In the NFL TE quality isn't as sparse, as there are typically just a few elite ones who keep performing at high levels. Another reason can be due to mostly all of their fantasy points being attributed to receiving statistics, being rather one dimensional. Lastly, there is usually just one TE on the field for a team, so the focus can be kept onto them. These reasons make them less volatile and easier to predict. WRs are similar to TEs where a majority of their points come from just receiving, but there are usually three on a field, which explains their slightly harder difficulty to predict. RBs gain their points from rushing but can also gain a lot from receiving, which can make it harder to predict dealing with more features. In a similar fashion, QBs can gain their points from multiple methods, passing and rushing, which makes it harder to predict. Additionally, QBs in real life football can be very erratic and impacted by injuries more. Our predictive accuracy for each NFL positional group models how the position exists within a real-life NFL structure.

Overall, our results show the difficulty of predicting fantasy points as the subject is very noisy. The points can be severely impacted by the unpredictability of the NFL, hurting the projections. However, the rankings of players in respect to others were better, as they seemed typical, made sense, and didn't raise any abnormalities we wouldn't have expected. Lastly, we did see our novelty implementation work, as we were able to produce predicted points and rankings for rookies, with the results being good. This helped prove our ideas of giving historical data to rookie players to help their predictive accuracy when they are rather noisy.

VI. CONCLUSIONS AND FUTURE WORK

This research's goal was to improve the already existing models that exist for predicting player fantasy football scores. The main goal for improvement was to fill a key gap in all of the prior research: the inability to predict rookie performances. To overcome this issue, we designed a model pipeline that separates rookies from veterans and feeds them into two separate models. This allowed for both groups to be modeled using features that were important to each group respectively. This process consisted of extensive data acquisition, cleaning, and feature engineering of both college and NFL statistics. Through these processes we were able to construct meaningful and consistent datasets for rookies and veterans in order to serve as a foundation supporting our machine learning models. The models were also sufficient in predicting both rookies and veterans. While the actual projection for points was a bit off the mark, our models made up for it in terms of numerical rankings. Our model was able to rank the players rather successfully based on real-world end of season rankings.

By bridging the gap between college and professional experience, our approach was able to demonstrate that rookie projection is actually possible, which is something the reference papers lacked. This was done by treating rookies as a structured modeling problem rather than just as an unpredictable outlier. Our modeling pipeline not only extends the work of the parent paper but also highlights the importance of non-quantitative statistics when it comes to forecasting sports environments. While our framework does accomplish the sought-out goals of including rookies and adding contextual features, there remain further opportunities to expand the work and strengthen the predictive power of our models.

A. Injury Detection for NFL Players

The first idea for improving upon our research is to implement a feature for injuries to players. Potential for injury detection for players could help improve the prediction of our model significantly. When looking at the RMSE for projected points, many of the contributors of high error were either players that were projected many points but then got injured early in the season; or they were back-up players that got opportunities due to injuries, but when the starters returned, they had a reduced role. The model also tended to underpredict players who were injured for the year before. The model also didn't account for interdependence between football positions. If one position is injured, the other position may suffer due to that (e.g. A WR may have a drop in their performance if their QB is injured).

This approach may provide significant improvement in our predictive accuracy. It also may offer other improvements beyond injuries including the positional interdependence that exist in football.

However, the actual implementation of this injury detection may be difficult. One potential source of injury data may come from looking at the history of past injuries to a player. This will be useful for detecting players that carry lingering injuries throughout their career, labelling them a injury prone. But this will not be helpful for predicting hidden injuries or freak accidents. Another approach may be to monitor biometric data that is collected by teams during practices and games. This approach may help to detect hidden injuries among players. But this approach is also not able to account for the freak accidents that may occur with the physical nature of football. This biometric data will also likely not be publicly available for anyone the use. Teams typically keep such data private out of respect for a player's privacy and also to keep sensitive information away from direct competition in the NFL.

If we figured out an effective way to implement this idea for injury detections, we believe that we could improve our projections for both veteran and rookie players. Our RMSE would be reduced significantly due to this approach reducing or completely eliminating high residual points in our dataset.

B. Draft Optimization

The next idea for future work is an approach that will optimize our player ranking based on an actual fantasy football draft. Our current approach focuses on just finding the highest scoring players, it ignores other factors that are at play when drafting a team to compete. It fails to account for the difference in league settings and formats. For example, certain league formats only allow one player of the QB position to start at a time. This seriously diminishes the value of drafting a QB with an early pick. Instead, more

focus should be placed on drafting other positions. To contrast, other leagues may allow two QBs to start at a time. This format typically drafts QBs in early round because of the high amount of points they can achieve. This goes hand in hand with other positional constraints that exist in fantasy league formats. Starting lineups may allow for more of one position than another. Our tool needs to be able to account for such constraints. Another thing we need to account for when optimizing is the decisions of fellow league mates.

This approach allows us to add real-world usefulness to our models. Implementing such an optimization feature like mentioned in [3] across an entire season, will allow fantasy football users to help improve their draft decisions to make their team better. The tool will also be able to account for the varying league settings, which can help all fantasy users not matter the presets.

For this approach there may be an issue of scalability, there are so many different types of fantasy league setups. There are many things that can change from league to league. League commissioners can edit the amount of fantasy managers in their league, the number of player per position group, and even the scoring settings. Our current model only takes scoring settings into account, where we use standard "Point-Per-Reception" (PPR). We do not account for other standard scoring settings or custom ones. The optimization model would also need to assume that each user in the league is drafting optimally. This assumption would need to be made if this optimization is done pre-draft. But the reality is that no player will draft completely optimally, there are always going to be some personal biases at play when drafting a fantasy football team. This leads into the next issue with this approach, there is no way that the approach can account for human bias present in the drafting process.

Should this optimization-based approach work, we believe that it will make the tool more user friendly overall. This would help fantasy managers draft the best team possible given their team and league constraints. This approach will help managers to make better decisions, maximizing the points for their team. Issues do arise through this approach, including the optimality assumptions that we need to have. Accounting for league mates' decisions would mean there would need to be a dynamic approach to predicting fantasy football.

C. Dynamic Approach to Projecting Fantasy Football

Our final alternate approach to extending our current research involves approaching fantasy football optimization as a ever-changing dynamic environment. Our current approach uses only the preseason data to project the season before it occurs. This way of approaching the problem may not fully capture a player's full potential within fantasy football. Since fantasy football is a year-round competition, a manager's roster will always be changing; it is very unlikely that a roster will remain unchanged throughout the season. This new approach would allow week-to-week insights for lineup construction. This can help with weekly decisions on what players to start or bench. It can also provide insights into picking players up off of waivers or free agency as well as trading players. This approach would constantly collect data about the season as it occurs, allowing for our models to get more recent key performance indicators.

With this new idea implemented, our tool could help users identify players that they should either trade for or away based on performance trends. It will also help users to pick up available players through waivers which often provide value in the future. This tool can also optimize starting lineups, considering matchup ratings and unforeseen injuries.

Unfortunately, this approach will involve a lot more data than the current model pipeline. It could prove difficult to collect such large amounts of data from every player across the league during every week of the season. This would require some kind of automation for extracting data from various sources. This would require advanced web-scraping or access to some kind of API access to pull our data. Assuming the data was easily accessible, another problem arises involving the intensity of analyzing this data. Such large amounts of data would need to be cleaned, normalized and features would need to be engineered. Processing the data is just the start, there is also a lot of computational intensity involved with training machine learning models. With such large amounts of data, this idea would require machines that have a lot more computational intensity than laptops.

If we were able to get past the issue of computational power, we think that this dynamic approach to modeling fantasy football will help improve on the base prediction that we made at the start of the season. We can use our current model to set a baseline for the upcoming season and then use this dynamic approach to incrementally improve upon those preseason predictions. This can help users make decisions during the season instead of just during the startup draft. Decisions can come in the form of trades, waiver pick-ups, and lineup optimization.

VII. REFERENCES

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[3] Mahoney, Joseph M, and Paniak, Tomasz B. "Method and Validation for Optimal Lineup Creation for Daily Fantasy Football Using Machine Learning and Linear Programming." *ArXiv.org*, 2024, arxiv.org/abs/2309.15253.